How to Install Visual Studio Code (Mac and Windows) - v2

Version 2. Screenshot captured 2026-11-10 from the official website; website menus and installer windows change, so follow the function of each button if a label differs. Scope: this guide installs the editor and checks that it runs a Python script. A short optional section at the end covers notebooks. It has no dependency on any workshop files.

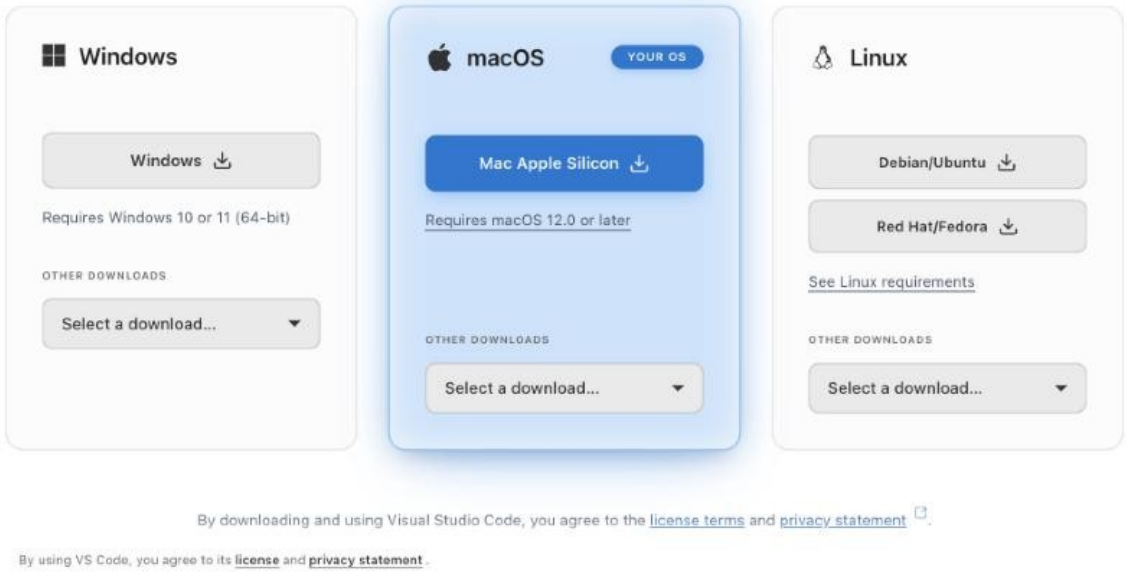
1. What is VS Code?

Visual Studio Code (VS Code) is a free code editor from Microsoft. You will use it to write and run Python files and, optionally, notebooks. It colours the code, helps you run it and includes a built-in terminal.

2. Requirements

- Windows: Windows 10 or 11, 64-bit.
- macOS: macOS 12.0 or later (Apple silicon and Intel Macs are both supported), as listed on the download page when this guide was written.
- About 500 MB free disk space and an internet connection for the download.
- On a work or school computer you may need an administrator to install software; the Windows "User setup" below does not need admin rights.

The official download page is code.visualstudio.com/download. It detects your operating system and highlights the matching button.



The VS Code download page (captured 2026-11-10): separate cards for Windows, macOS and Linux, each with an extra menu for other builds. The wording of that menu changes from time to time (for example "Select a download", "Other builds" or "All downloads"); look for the link that lists other installers, ZIP files and processor types.

3. Install on macOS

1. Open code.visualstudio.com/download and click the macOS button. If your Mac has an Intel chip, or you are not sure, open the menu of other builds under macOS and choose the Intel or Universal build. To check your chip: Apple menu > About This Mac.

2. Open the downloaded file (a .dmg or .zip). If you get a .zip, double-click it to unpack Visual Studio Code.app.

3. Drag Visual Studio Code.app into your Applications folder.

4. Open VS Code from Applications (or search with Spotlight, Cmd+Space). macOS may ask whether you are sure you want to open an app downloaded from the internet; click Open.

5. Optional: Control-click the VS Code icon in the Dock and choose Options > Keep in Dock.

Make the "code" command work in the Terminal (optional but useful)

Use the Command Palette method first. It is the most reliable.

1. In VS Code press Cmd+Shift+P to open the Command Palette.

2. Type "Shell Command" and choose: Shell Command: Install 'code' and optionally 'copilot' commands in PATH. If asked about installing GitHub Copilot CLI you can choose No; it is not needed.

3. Close and reopen your Terminal, then run the command below inside any folder to open that folder in VS Code.

```
code .
```

Only if that menu item is missing, you can add VS Code to your PATH by hand. These instructions are for the zsh shell (the default on current Macs). Add the following ONE line to the file ~/.zprofile, then open a new Terminal. Copy it as a single line; if your PDF viewer splits it, join it back together so that "Visual Studio Code.app" stays on one line with a single space between the words.

```
export PATH="$PATH:/Applications/Visual Studio Code.app/Contents/Resources/app/bin"
```

4. Install on Windows

Choose one of two installers. For most people, the User setup is best.

- User setup (recommended): no administrator rights needed; installs only for you in your user folder and updates itself smoothly.
- System setup: needs administrator rights; installs in Program Files for all users on the computer.

1. Open code.visualstudio.com/download and click the Windows button. For the System setup, or for ARM or ZIP versions, use the menu of other builds under Windows.

2. Open the downloaded file (named like VSCodeUserSetup-<version>.exe). If Windows asks "Do you want to allow this app to make changes?", click Yes.

3. Accept the licence agreement and click Next through the pages. Keep the default install location.

4. On the options page, leave the default options checked. If you see Add to PATH, keep it checked. You may also tick "Add Open with Code action" to right-click files and folders. If a Copilot CLI page appears, you can choose to skip it.

5. Click Install, then Finish. Leave "Launch Visual Studio Code" checked to open it.

If the installer is blocked by your computer, the website offers a ZIP version: unzip it into a folder in your user area (for example C:\Users\<you>\AppData\Local\Programs\VS Code) and run Code.exe. The ZIP does not update itself.

5. Check that it works

1. Open VS Code. A Welcome page appears. It may offer to sign in to GitHub Copilot; this is optional, and you can choose "Continue without Signing In".

2. Open a terminal (macOS Terminal; on Windows, Command Prompt or PowerShell) and run the command below. It should print a version number.

```
code --version
```

If the command is not found on Windows, close and reopen the terminal (or sign out and back in) so it picks up the new PATH. On macOS, use the Command Palette step above.

6. Add Python support and run a script

You also need Python installed on your computer; see the separate Python guide.

1. Open the Extensions view (the square icon in the left bar), search for Python and install the one published by Microsoft.

2. Choose File > New Text File, type the line below, and save it as hello.py in a folder you can find.

3. Select your Python: Command Palette (Cmd/Ctrl+Shift+P) > Python: Select Interpreter, then pick the Python you installed (or your .venv, see the Python guide).

4. Click the Run triangle at the top right of the editor. The output "Hello, bioinformatics!" appears in the terminal panel.

```
print("Hello, bioinformatics!")
```

7. Optional: notebooks in VS Code

Only needed if your instructor asks you to run .ipynb notebooks on your own computer. Google Colab needs none of this.

1. Install the Jupyter extension (also published by Microsoft) from the Extensions view.

2. Create a file with File > New File > Jupyter Notebook, or open an existing .ipynb file.

3. Click Select Kernel at the top right and choose the Python environment you selected above. If VS Code offers to install ipykernel into that environment, accept.

4. Type the print line above into a cell and run it with the play button or Shift+Enter.

8. Troubleshooting

- macOS says the app can not be opened: you may have opened it from the .dmg or Downloads folder; move it to Applications first, then Control-click and choose Open.
- Windows says you need administrator rights: use the User setup, or ask your IT department.
- Installer download is very slow or blocked: use the ZIP version or download on another network.

- "code" command not found: reopen the terminal; on macOS use the Command Palette step above.
- Run button does nothing or says no interpreter: use Python: Select Interpreter.
- Update later: VS Code shows an Update button when a new version is available (macOS: restart; Windows User setup: update in place).